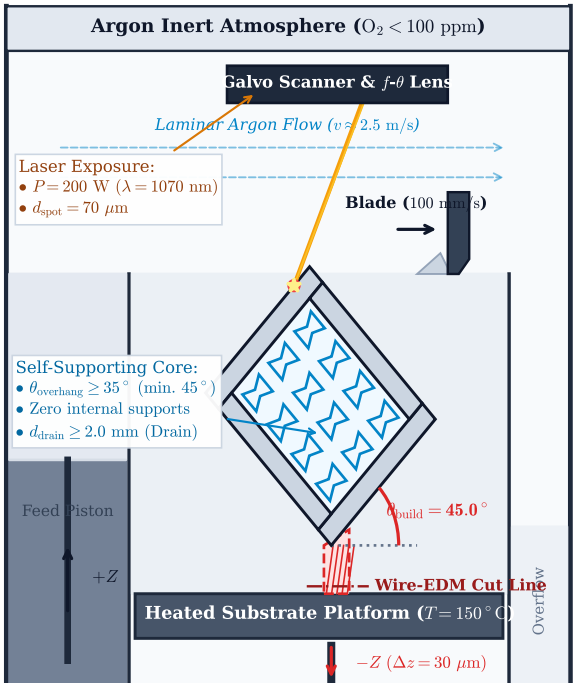
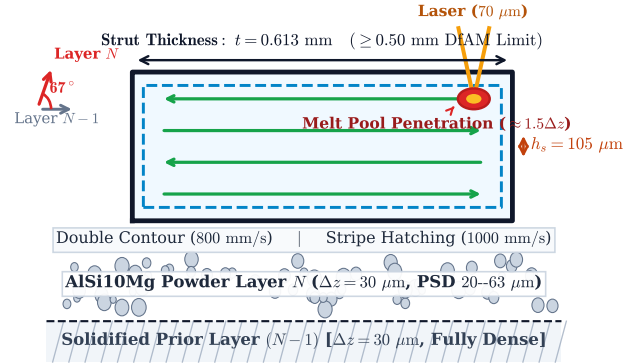
(a) L-PBF Build Chamber & 45° Orientation



(b) Laser Scan Strategy & Process Window



L-PBF CERTIFIED PROCESS PARAMETERS (EOS M 290 / AlSi10Mg)

Laser Exposure & Energy	Hatching & Densification
• Laser Power: $P = 200$ W (Yb-fiber)	• Hatch Spacing: $h_s = 105$ μm
• Hatch Velocity: $v = 1000$ mm/s	• Interlayer Rotation: $\Delta\alpha = 67.0^\circ$
• Contour Velocity: $v_c = 800$ mm/s	• Volumetric Energy: $\mathbf{E_v = 63.49\ J/mm^3}$
• Spot Diameter: $d_{spot} = 70$ μm	• Relative Density: $>$ **99.8**% (Dense)
• Layer Height: $\Delta z = 30$ μm	• Platform Preheat: $T_p = 150^\circ$C